

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456772.67042 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.023
Transit depth (Rp/Rs)^2	1.65 +/- 0.59 [%]
Orbital Inclination (inc)	82.06 +/- 1.71
Airmass coefficient 1 (a1)	1.168 +/- 0.011
Airmass coefficient 2 (a2)	-0.104 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	None
Optimal Aperture	4.1
Optimal Annulus	15.91
Transit Duration (day)	0.0481 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.128 ± 0.023 vs 0.1594 (deviation 1.28σ)
Duration fit vs published	0.0481 d vs 0.0512 d (deviation 0.49σ)
Mid-time O-C	-0.2 min
Depth SNR	5.2
Residual scatter	0.94 %

Light curve

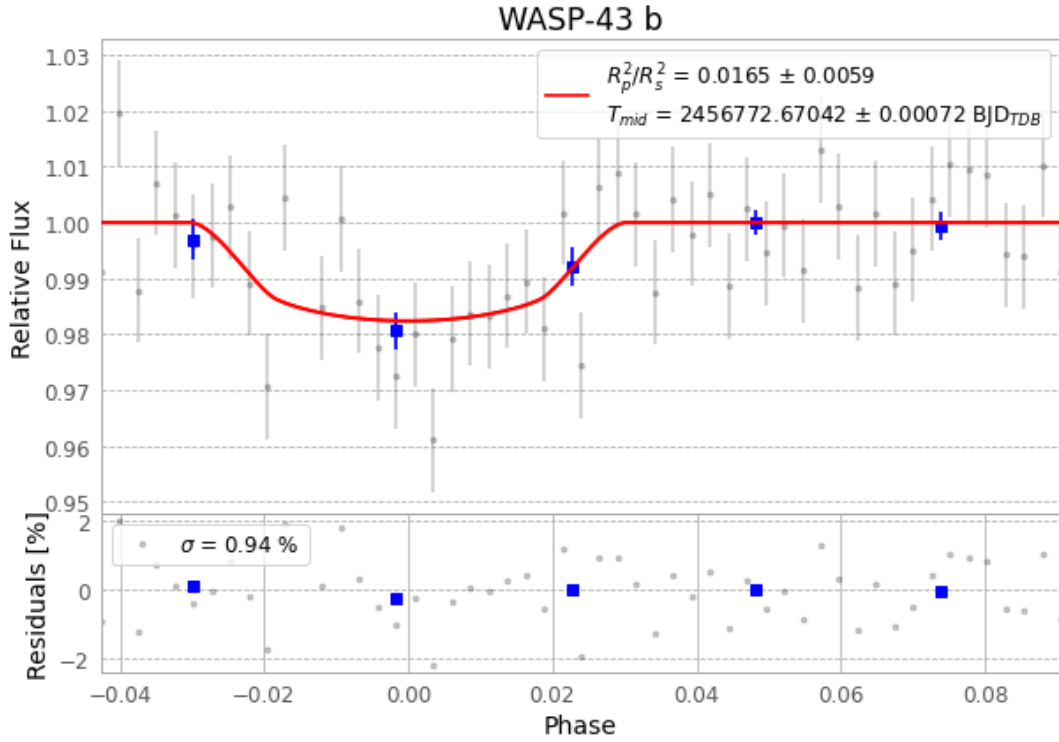


Figure 1 — FinalLightCurve_WASP-43 b_2014-04-24.png (SHA-256 498ee44a9afc6c10...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 569ea5a4dd972bcb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), WASP-43140425030913.FITS → WASP-43140425054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140425.log (720ce8ad811b...), FinalLightCurve_WASP-43 b_2014-04-24.png (498ee44a9afc...), FinalLightCurve_WASP-43 b_2014-04-24.csv (0b8b2c1b595a...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:16Z.